

From Representation to Realization: What an Outcome Does Not Tell You About Who Caused It

Human-readable companion to From Representation to Realization: Stage-Resolved Causal Attribution in Apparently Agentic Transitions

Human-readable companion to the completed SRBS manuscript. This version preserves the scientific claims and limits of the submitted manuscript while presenting the motivating problem and attribution architecture in more direct language. It does not constitute a separate research contribution.

Abstract

An observed action can look as though one focal system represented an option, chose it, implemented it, and caused the final outcome. The same endpoint, however, can be produced by different causal organizations. A represented alternative may never participate causally in selection. A participating alternative may lose. A focal state may genuinely affect selection while that state itself was prepared by nonfocal causes. A selected state may fail to produce focal implementation, and a realized outcome may depend on assistance, prevention, or bypass outside the focal subsystem. This paper therefore treats apparently agentic behavior as a causal-attribution problem rather than as evidence of a unitary controller. It separates three local identification problems: candidate participation (Test -1), selection provenance (Test 0), and realization provenance (Test 1). Stronger end-to-end attribution is licensed only when the required propositions remain valid across their interfaces and when any causal relation carried from one experimental regime to another has an independently justified validity basis. The protocol uses existing causal-inference, mediation, transportability, causal-abstraction, representation-intervention, action-selection, and execution machinery. Its contribution is the discipline of composing those local results without erasing nonfocal preparation, bypass, joint contribution, or unresolved provenance.

1. The everyday mistake: reading a whole causal story from one successful outcome

Suppose a system ends by doing A. Perhaps A was one of several represented possibilities, and the final movement, choice, or transition looks exactly like the system selected A and carried it out. It is tempting to tell one continuous story: A was represented, A entered the competition, the focal subsystem selected it, the focal subsystem implemented it, and the observed world changed because of that implementation.

The SRBS Endgame manuscript begins by refusing that shortcut. The same observed outcome can remain compatible with several upstream causal organizations. A may have been represented but causally irrelevant to selection. It may have participated and lost while some later process still produced an A-like outcome. A focal state may have contributed causally to selection while the state itself was prepared by a nonfocal variable. A selected state may never be focally implemented. Or focal implementation may occur while the final environmental outcome depends on additional nonfocal support. Conventional action science already distinguishes stages and components of action specification, selection, implementation, and effects; Endgame's narrower question is what causal attribution the available evidence actually licenses (Cisek, 2007; Brass & Haggard, 2008; Pacherie, 2008; Chambon et al., 2014).

observed outcome $\neq$ complete causal provenance

2. Boundary first: 'focal' is a query-relative attribution role

The protocol uses an analytical boundary B to separate variables treated as focal from variables treated as nonfocal for a particular causal-provenance question. This is a methodological boundary, not a metaphysical claim that the focal system is an independently existing agent. Changing B changes which provenance questions can be expressed and tested.

That boundary discipline matters because finding an upstream nonfocal cause of a focal state does not erase a genuine downstream effect of that focal state. If intervention on focal state z changes selected state D , then $z \rightarrow D$ can remain a valid local causal relation even after a larger analysis discovers $x \rightarrow z$. The correct record becomes $x \rightarrow z \rightarrow D$. One fact refines provenance; it does not invalidate the other.

The manuscript therefore limits its strongest conclusions to bounded attribution: causal claims are relative to the modeled variables, query, observation family, intervention regime, and tested nonfocal set. No finite model establishes the absence of every conceivable upstream cause outside the chosen boundary.

3. Three identification failures that an endpoint cannot repair

Failure	What may be known	What remains unidentified
Representation is not participation	Alternative A is retained, retrievable, reportable, decodable, or otherwise representationally available.	Whether A currently exerts a causal effect on the dynamics relevant to selection.
Participation is not selection provenance	A and B both causally participate in the candidate dynamics.	Whether the selected state D was causally supplied by a focal state, a nonfocal state, a nonfocally prepared focal state, or joint contribution.
Selection is not implementation or realization	$D=A$ is independently established.	Whether A is focally implemented (EF), and whether that implementation causally produces the realized transition Y rather than being prevented, assisted, redirected, or bypassed.

These are attribution propositions, not mandatory biological stages. Their logical order tells us which proposition must be supported before a stronger end-to-end claim can be promoted. The underlying physical dynamics may be parallel, recurrent, distributed, overlapping, graded, or continuously evolving.

4. Test -1: did the represented possibility actually participate?

Test -1 begins with representational availability, written in the manuscript as $MR(A)=1$. That is only the permission to ask a participation question. The test then uses a candidate-specific experimentally addressable state qA and a candidate-sensitive preselection target ZC . The candidate set C_t contains alternatives such as A; qA is evidence used to test A, not a silent redefinition of A.

A candidate-participation certificate requires four things:

- A representational criterion for A that is independent of the participation effect being tested.
- A candidate-specific variable or state qA that addresses A rather than indiscriminately perturbing the selection system.
- A component of ZC whose causal relevance to the prespecified selection-attribution query has been independently established in the inference regime, or validly carried into it.
- Under the prespecified candidate-specific intervention family, changing qA changes that validated selection-relevant component of ZC while the representational and qualification conditions required by the contrast remain satisfied.

This prevents circularity. Participation is not defined as 'whatever makes some preselection signal change.'

Prediction, decoding, salience, attention, arousal, conflict, generic readiness, or generic network influence is not enough by itself.

The hardest limit is that a perturbation can both reveal and create the causal leverage being attributed. Presenting A again, increasing its salience, or stimulating a latent representation may recruit or reactivate it. Perturbational working-memory research makes that possibility concrete (Rose et al., 2016; Wolff et al., 2017). If no design can distinguish pre-existing participation from intervention-induced participation with adequate construct validity, the stronger Test -1 attribution remains unidentified.

5. Test 0: what caused the selected state?

For the manuscript's complete end-to-end query, Test 0 is interpreted as selection provenance only after the relevant candidates have passed Test -1 in the regime needed for that attribution. The selected state D must also be identified independently of the final outcome. An A-like outcome cannot retrospectively certify $D=A$ because later implementation, assistance, bypass, or environmental dynamics could generate the same endpoint.

Let zD be a focal state hypothesized to contribute to D and $x0$ a relevant nonfocal selection-stage state. Ordinary structural-causal, mediation, path-specific, or related inherited tools can then test several provenance relations. The manuscript is careful not to force them into mutually exclusive mechanism boxes.

Selection-provenance pattern	Licensed interpretation
$zD \rightarrow D$	A focal state contributes causally to the independently identified selected state.
$x0 \rightarrow D$	A nonfocal state contributes causally to selection.
$x0 \rightarrow zD \rightarrow D$	A nonfocal state prepares a focal state that itself remains causally effective.
$zD \rightarrow D$ and $x0 \rightarrow D$	Joint focal-nonfocal contribution; interaction or synergy is not inferred unless separately identified.

A critical composition rule already present in the submitted SRBS manuscript is that the candidate-participation premise required by Test 0 must remain valid in the regime where the Test-0 result is used. If a Test-0 manipulation removes A from the validated candidate set, that condition no longer answers the intended selection-provenance question among the same participating candidates. The result may still be scientifically useful, but it cannot repair the missing premise.

6. Test 1: selection, focal implementation, and realized outcome are different

Test 1 begins, for the complete attribution, only after $D=A$ has been independently identified in the relevant regime. It then separates the selected state from focal implementation and, where the scientific question requires it, separates focal implementation from the realized transition.

EF denotes an independently validated focal implementation outcome. It is a query role, not a universal biological state. If the target outcome is intentionally defined at the same boundary and level as focal implementation, EF and Y may merge and the implementation-versus-realization distinction disappears. If the scientific question keeps them separate, the implementation observable must remain independent.

A causal relation $D \rightarrow EF$ needs a valid causal-identification strategy; ordinary association between selection and implementation is not enough. Likewise, $EF \rightarrow Y$ can coexist with nonfocal prevention, preparation, assistance, or bypass. If $x1$ prepares EF and EF then genuinely affects Y, both $x1 \rightarrow EF$ and $EF \rightarrow Y$ remain in the account. If $x1$ independently produces Y, the outcome can match A even without the focal route required by the stronger attribution.

7. End-to-end attribution is composition of propositions, not concatenation of arrows

The three local tests answer different questions. Putting three locally correct results beside one another does not automatically create one globally valid causal story. Endgame therefore composes certified causal propositions and makes the interfaces explicit.

candidate participation $\Rightarrow$ selection provenance $\Rightarrow$ realization provenance

The double-arrow idea here is evidential: downstream attribution presupposes an upstream certificate or a validly carried equivalent. It is not a claim that biology runs through three discrete modules in that order.

The manuscript represents a generic validity record as $H=(P, R_s, R_t, V)$: the proposition P, the source regime R_s where it was established, the target regime R_t where it is needed, and the inherited validity basis V for carrying it. The proposition can be revalidated in the target regime or supported by ordinary invariance, transportability, causal-transformation, or abstraction machinery where those assumptions are justified (Pearl & Bareinboim, 2011, 2014; Rubenstein et al., 2017; Beckers & Halpern, 2019; Rischel & Weichwald, 2021; Xia & Bareinboim, 2025).

Two kinds of validity record must not be confused. An inter-stage prerequisite handoff carries a premise such as candidate participation into Test 0 or $D=A$ into Test 1. A causal relation estimated inside one stage can separately require source-target validation if it was learned under a different regime. One record cannot silently do the other's job.

8. Why a correct local intervention can still fail the global attribution

The manuscript's worked counterexample makes this concrete. Imagine two systems that have the same candidate-participation and selection certificates in the target regime, including $D=A$. In a separate realization-test source regime, both obey $EF:=D$, so the same manipulation correctly identifies a local $D \rightarrow EF$ relation.

In one system, that relation remains valid in the target regime and Y depends on EF . In the other, the target regime changes: EF is fixed away from D while a nonfocal x_1 produces Y . The observed target outcome can still be A -like, and the source experiment was not wrong. What fails is the promotion of that locally correct source-regime relation into the target-regime end-to-end attribution.

The lesson is not that interventions are untrustworthy. It is that the proposition being transported has to be named and its validity justified. A successful causal estimate in one regime does not become a universal premise merely because a later outcome looks compatible with it.

9. The observable roles are query-dependent and may legitimately merge

The protocol does not claim that every experiment needs a fixed universal inventory of variables. MR is needed only if the scientific query distinguishes representational absence from representation without participation. ZC is needed when candidate participation must be distinguished from losing competition. D is needed whenever selection must be identified independently of downstream implementation or realization. EF is needed only if implementation and realization are scientifically distinct.

Variables may merge, but only after causal validation or after deliberately narrowing the question. ZC and zD can be one measured state only if that state independently satisfies both roles. D and EF can merge if one empirical state is independently validated for both selection and implementation. EF and Y can merge if the outcome is intentionally defined at the focal implementation boundary. Correlation, shared anatomy, temporal proximity, or mutual predictability is not enough.

The same parsimony rule applies to history. If otherwise matched present states respond differently under the relevant intervention family, the present-state description is insufficient for that query. But if an augmented present state restores the relevant sufficiency, the protocol does not preserve raw history as an irreducible causal mechanism.

10. A concrete scientific mapping: constrained reach selection

The SRBS manuscript uses constrained reaching as an illustrative mapping, not as evidence that any cited study instantiated the whole protocol. Dorsal premotor activity can represent multiple potential reach directions before later differentiation toward a selected reach (Cisek & Kalaska, 2005). That makes the domain useful for showing how representation, selection, preparation, implementation, and movement can be measured separately.

A direction-specific signal associated with A could supply a candidate-specific qA , but decoding that signal would establish availability or specification rather than participation. A preselection target counts as ZC only if the particular component used for the inference has independently established causal relevance to the reach-selection query. $D=A$ would require an independently validated selection criterion before overt execution. EF could be a validated target-specific descending motor or EMG outcome; the external hand or cursor transition would be Y .

Preparation and movement-generating activity need not coincide (Kaufman et al., 2014; Kim et al., 2021), and perturbation-to-behavior inference can be affected by network propagation, state dependence, and construct validity (Bergmann & Hartwigsen, 2021). The example therefore demonstrates the protocol's discipline: a missing certificate or validity basis remains missing rather than being supplied by the fact that the movement eventually occurred.

11. The protocol has explicit stopping points

Missing element	What must remain unresolved
Independent representation measure, when the query needs it	Representational absence versus representation without participation.
Validated selection-relevant ZC or a valid basis into the inference regime	Whether a qA effect certifies selection-relevant candidate participation.
Candidate-specific causal identification	Representation may be established while Test -1 participation remains unidentified.
Independent D	Selection versus downstream implementation or realization.
Causal identification of zD or relevant x0	Selection-provenance alternatives.
Independent EF when implementation is distinct from Y	Focal implementation failure versus downstream failure, assistance, or bypass.
Valid D → EF identification	Selection-implementation association cannot be promoted to causation.
Relevant nonfocal variation	Local focal contribution may stand, but stronger exclusion of preparation, bypass, or joint contribution cannot.
Certified source-target validity	A proposition learned in one regime cannot be used as a downstream premise in another merely because both local analyses are correct.

Promotion of the manuscript's complete end-to-end attribution stops at the first missing required proposition or validity condition. Later-stage local relations may still be scientifically studied and identified independently. The protocol therefore turns uncertainty into an explicit bounded result rather than forcing a full causal story.

12. Four levels of warranted focal attribution

Level	What the evidence supports	What it still does not support
1. Local focal causal participation	A focal variable contributes causally to one local target, such as $zD \rightarrow D$ or $EF \rightarrow Y$.	Complete end-to-end focal attribution or focal origin.
2. Candidate participation plus focal contribution across identified selection and realization stages	Test -1 is certified; the relevant focal contributions in Tests 0 and 1 and their required handoffs/validity conditions are identified.	Focal provenance of candidate participation, origin of focal states, or exclusion of untested nonfocal causes.
3. Focal contribution surviving tested nonfocal controls	Level 2 plus tested selection- and realization-stage preparation, bypass, and joint-contribution alternatives.	Absence of untested upstream causes or unrestricted autonomy.
4. Focal realization under defined external reference conditions	All lower certificates plus maintenance of the target transition under prespecified tested external reference/withdrawal conditions.	Technical or token sufficiency, self-causation, metaphysical independence, or absence of every external antecedent.

The phrase 'the focal system controlled the transition' is therefore scientifically underspecified unless the intended evidential level is stated. A valid local focal effect, a bounded end-to-end attribution, and an unrestricted claim of autonomy are not interchangeable conclusions.

13. Limits the protocol deliberately does not solve

First, the open-boundary limit remains. A tested model can identify focal contributions relative to a defined boundary and still omit upstream causes outside that model. Endgame does not turn finite causal analysis into proof of unrestricted causal origin.

Second, the local tests establish interventional causal contribution at the level supported by their design. They do not automatically establish that one variable was necessary or sufficient for the particular realized token. Actual-causation and probabilities-of-causation frameworks require additional counterfactual assumptions or evidence for those stronger claims (Pearl, 1999; Halpern, 2016; Li & Pearl, 2023).

Third, conceptual identifiability is not technological availability. Candidate-specific perturbation, independent selection measurement, focal implementation measurement, and valid source-target transport may be unavailable in a given domain. That does not make the causal distinction unreal; it means the attribution remains unidentified for that domain and query.

14. What the paper contributes - and what it does not

The causal machinery is inherited. Structural causal models provide intervention semantics; mediation and path-specific-effect tools provide route-specific causal questions; transportability and causal-abstraction work provide source-target and cross-description validity machinery; interventionally validated representations show how internal variables can be tested causally; and action science already supplies nonunitary selection and implementation structure.

The residual contribution is architectural. Endgame requires those inherited causal questions to be answered at the attribution stage where they matter, requires explicit certification when their results are carried across interfaces or regimes, and preserves nonfocal preparation, bypass, and joint contribution rather than upgrading local focal effects into global autonomy. The SRBS hostile review accepted that contribution as a coherent methodological systems paper while noting that its novelty is architectural rather than mathematical.

The protocol is therefore not a definition of agency, not a privileged controller theory, not a new causal calculus, and not a new action mechanism. It is a disciplined answer to a narrower question: given an apparently agentic transition, how far can the focal causal attribution legitimately go before the evidence runs out?

15. Conclusion

An outcome can be perfectly real and still be causally ambiguous. The final transition does not tell us whether a represented alternative participated in selection, what caused the selected state, whether the selected state was focally implemented, or whether focal implementation caused the final realized transition rather than being assisted, prepared, prevented, redirected, or bypassed.

Endgame resolves that ambiguity by refusing to let one stage certify another. Test -1 asks whether a represented alternative causally participates. Test 0 asks what contributes to the independently identified selected state. Test 1 asks how that selected state relates to focal implementation and realized outcome. The resulting propositions compose only when their prerequisites and source-target validity conditions remain justified.

The endpoint is therefore bounded by design. Stronger focal attribution is earned one proposition at a time, and it stops at the first unsupported proposition. A local causal effect remains real even when its state has nonfocal provenance; a complete focal story remains unwarranted when the necessary evidence is missing. That is the manuscript's core methodological result.

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